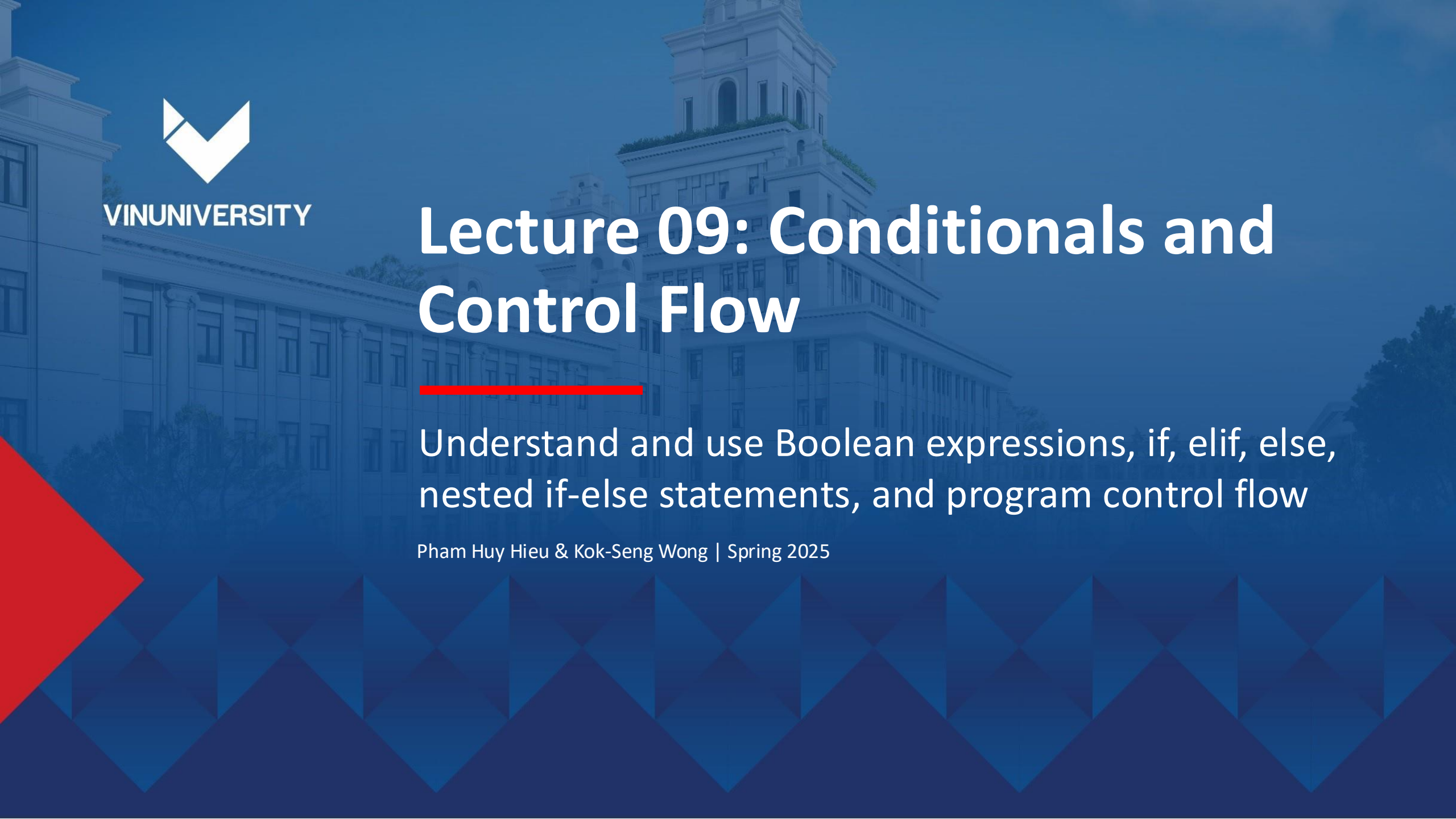


VINUNIVERSITY

Lecture 09: Conditionals and Control Flow

Understand and use Boolean expressions, if, elif, else, nested if-else statements, and program control flow

Pham Huy Hieu & Kok-Seng Wong | Spring 2025

COURSE ANNOUCEMENT

📢 Announcement to Students (Week 8 Schedule)

 November 4: Theory Exam
 November 6 & 7: Practical Exam

- Content: Week 01 to Week 07
- Students are advised to check the detailed exam schedule on the Syllabus for their class.

MOTIVATIONAL EXAMPLE 1

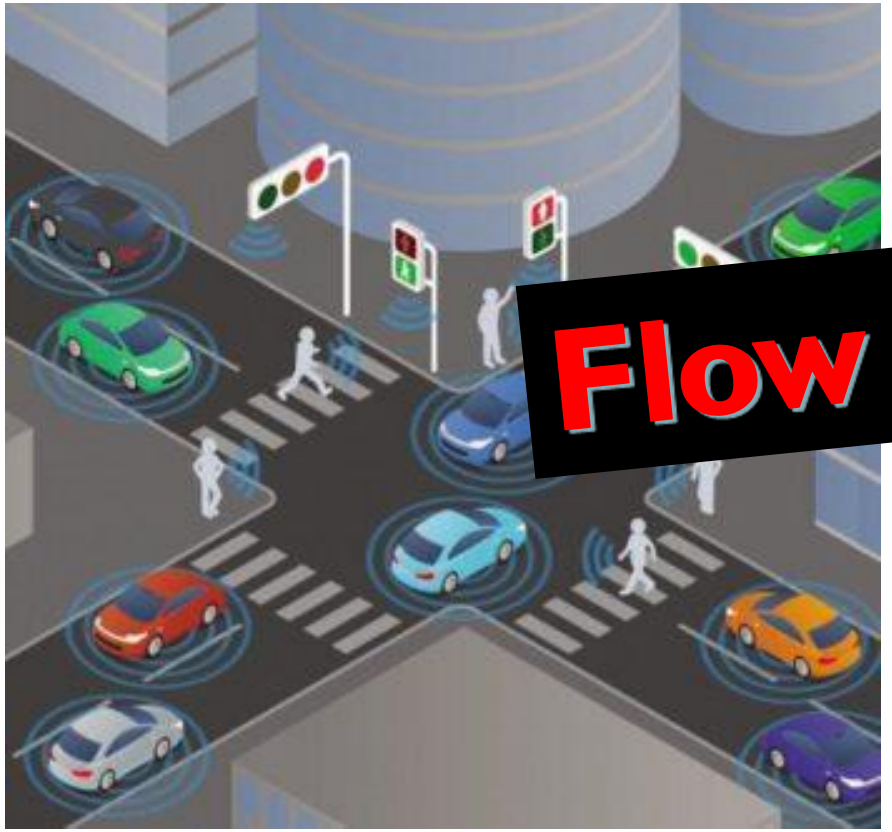


Ali Baba's mom said: "Yah, please go to Winmart and buy one bottle of milk. If they have eggs, bring 6."

Ali Baba came home with?

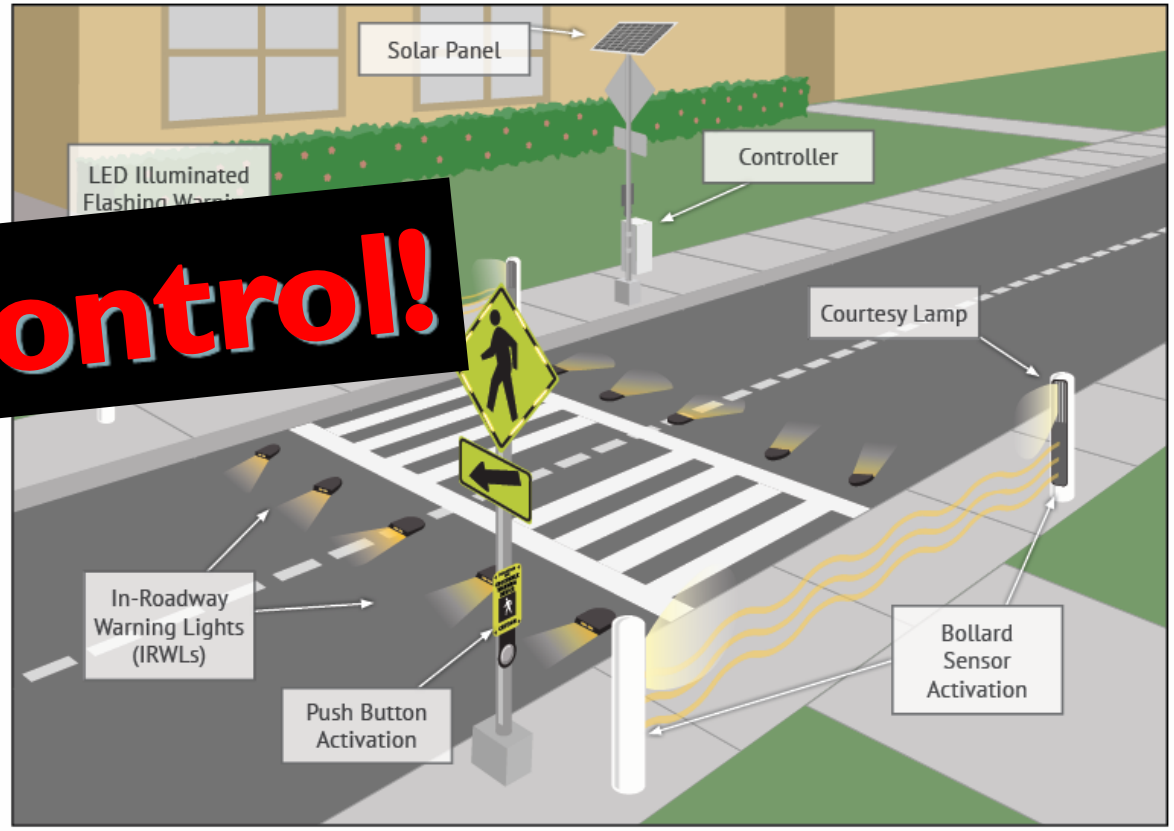
MOTIVATIONAL EXAMPLE 2

- Traffic light system



[source]

- Smart Crosswalk



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Flow Control!

LEARNING OUTCOMES

Upon the completion of this lecture, students will be able to:

- describe the syntax for conditional statements in Python.
- understand the control flow of if, else, elif, and nested if-else statements.
- write conditional statements in Python to control the flow of code.

(MATHEMATICS) LOGICAL CONDITIONS

- Equals: `a == b`
- Not Equals: `a != b`
- Less than: `a < b`
- Less than or equal to: `a <= b`
- Greater than: `a > b`
- Greater than or equal to: `a >= b`

CONDITIONALS: IF-STATEMENTS

- Format:

condition

```
if<boolean-expression>:
    <statement>
    …
    <statement>
<following_statement>
```

Second line must be indented

The colon (:) separates the header of the compound statement from the body.

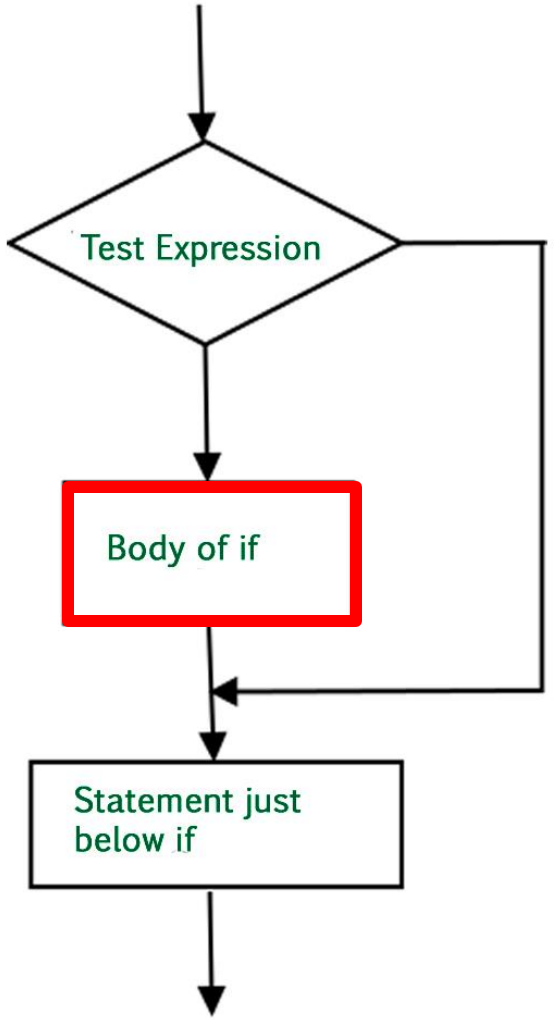
- If<*boolean-expression*> is true, then execute all of the statements indented directly underneath (until first non-indented statement).

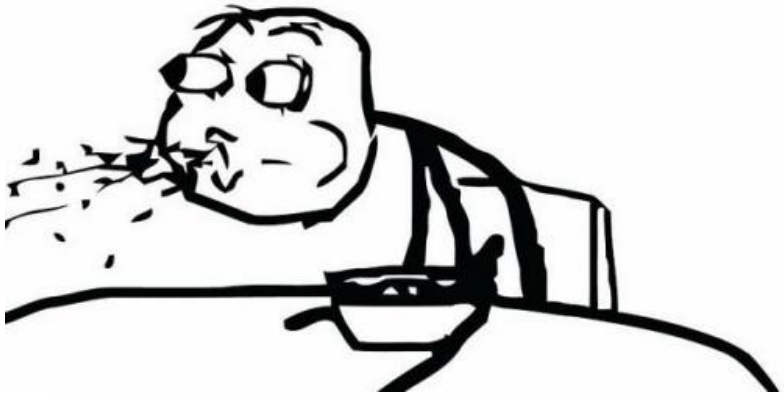
CONDITIONALS: IF-STATEMENTS

- Format: (other programming languages)

```
if<boolean-expression>{
    <statement>
    …
    <statement>
}
<following_statement>
```

CONTROL FLOW: IF-STATEMENT





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BOOLEAN EXPRESSIONS

- Boolean expressions are used to evaluate to a Boolean value (True or False).

```
is_raining = False
is_sunny = True
temperature = 32
```

Boolean variables:

```
if is_raining:
    print("You need an umbrella!")
```

Boolean comparisons:

```
if temperature > 28:
    print("Nice temperature!")
```

Boolean operations:

```
if not is_raining and is_sunny:
    print("Let's go out for a walk!")
```

```
if temperature > 30 or is_sunny:
    print("You should stay at home!")
```

CLASS PRACTICE

Sample Code:

A

```
a = 1
print(a)
```

B

```
a = 1
a += 1
print(a)
```

C

```
a = 1
if a == 1:
    a += 2
    print(a)
```

D

```
a = 1
if a == 2:
    a += 3
print(a)
```

E

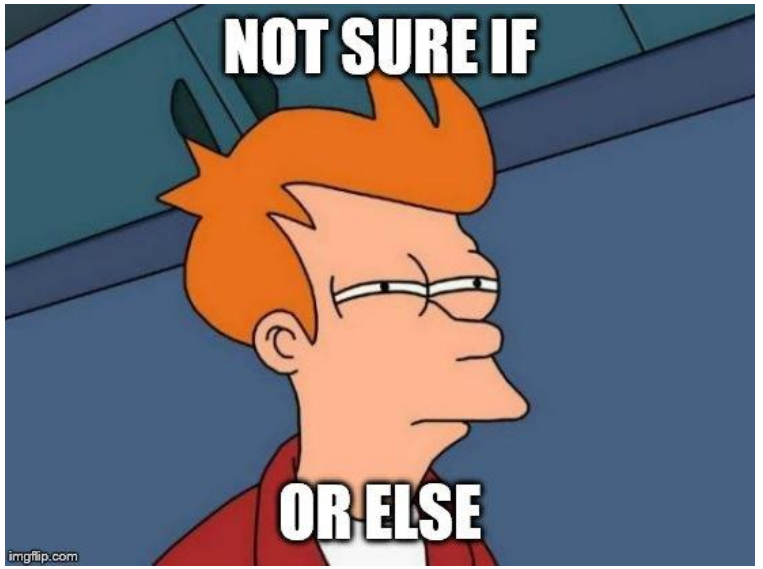
```
a = 1
if a == 2:
    a += 3
a += 4
print(a)
```

Output:

CONDITIONALS: IF-ELSE-STATEMENTS

- Format:

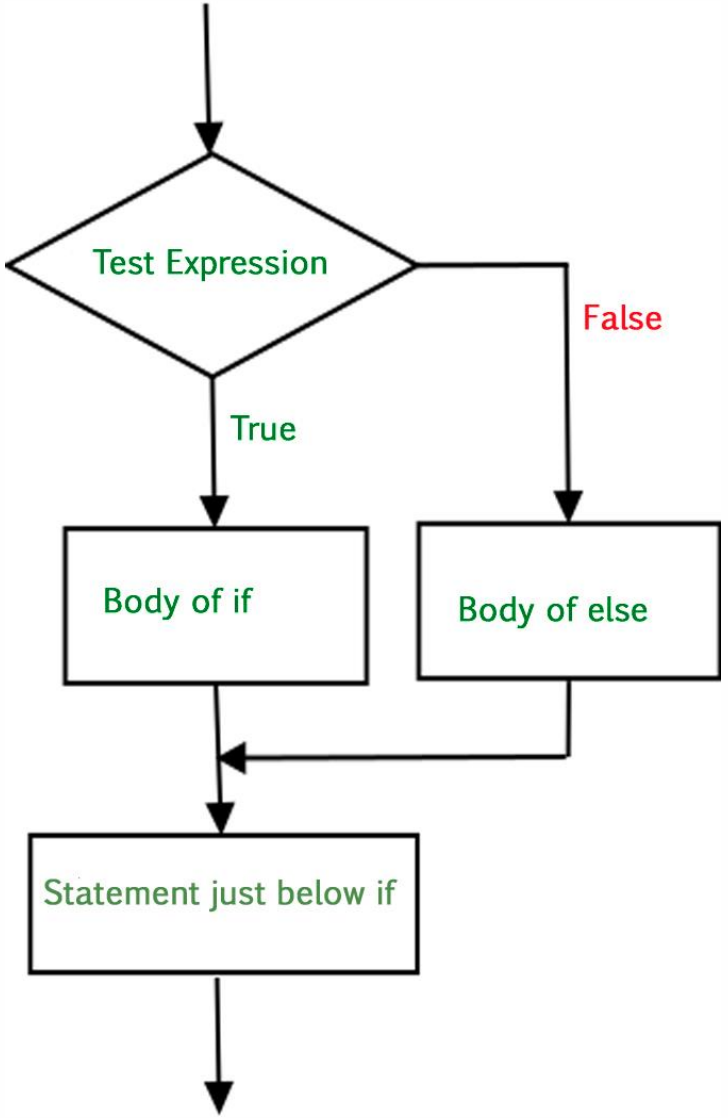
```
if<boolean-expression>:
    <statement>
    …
else:
    <statement>
    …
```



- if<*boolean-expression*> is true, then execute statements indented under `if`; otherwise, execute the statements indented under `else`

CONTROL FLOW: IF-ELSE

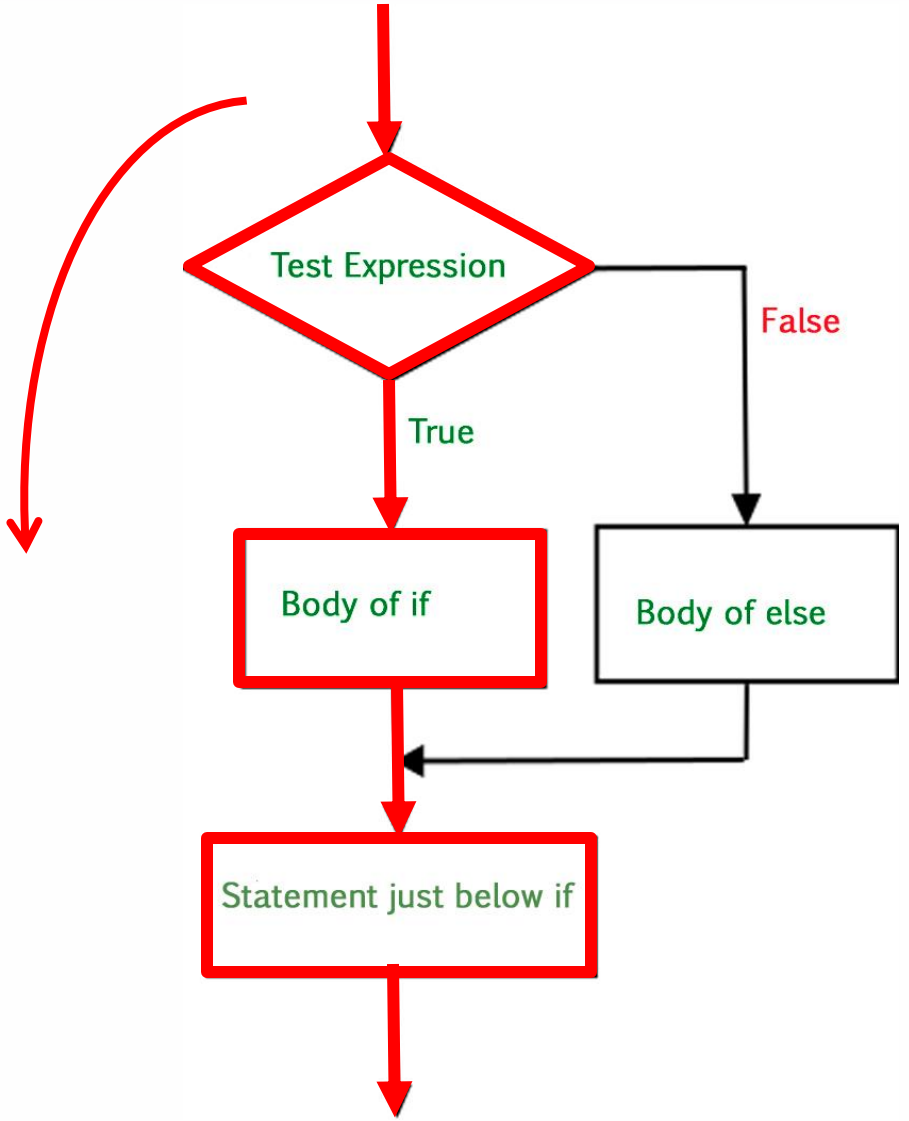
Flow: program only takes on path during an execution



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CONTROL FLOW: IF-ELSE

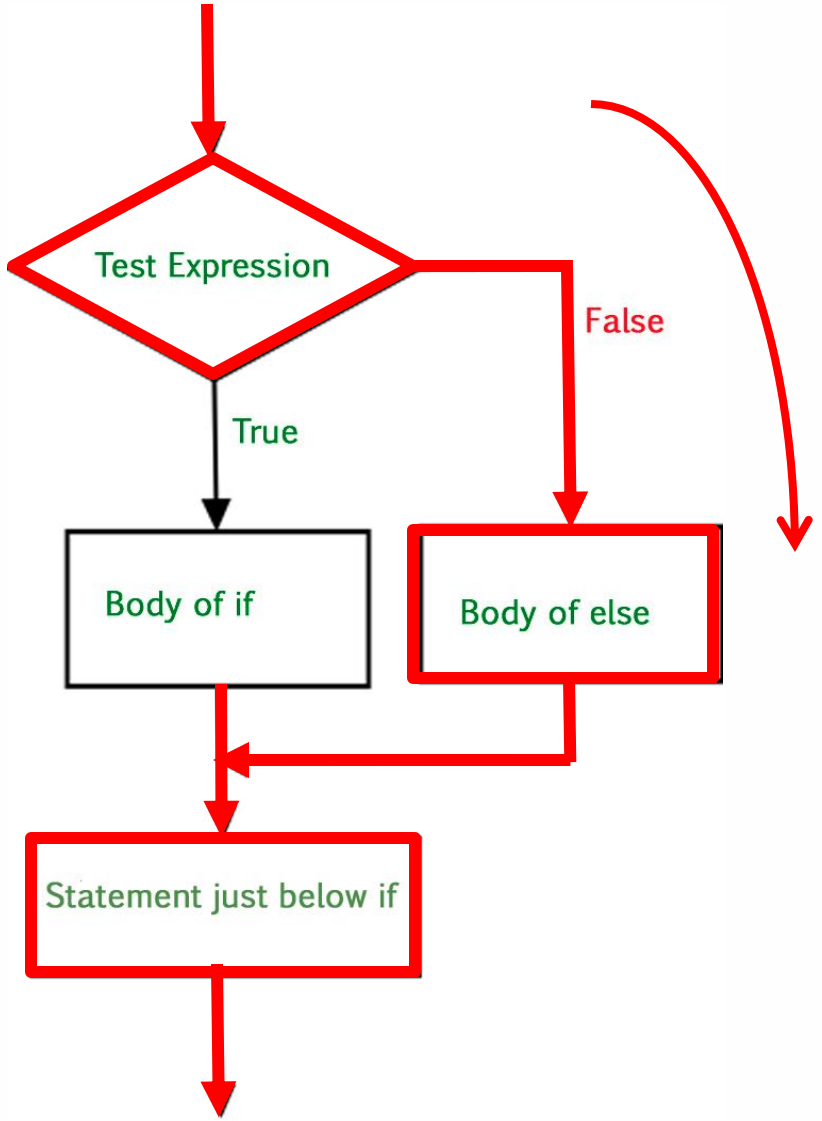
Flow: program only takes on path during an execution



[source]

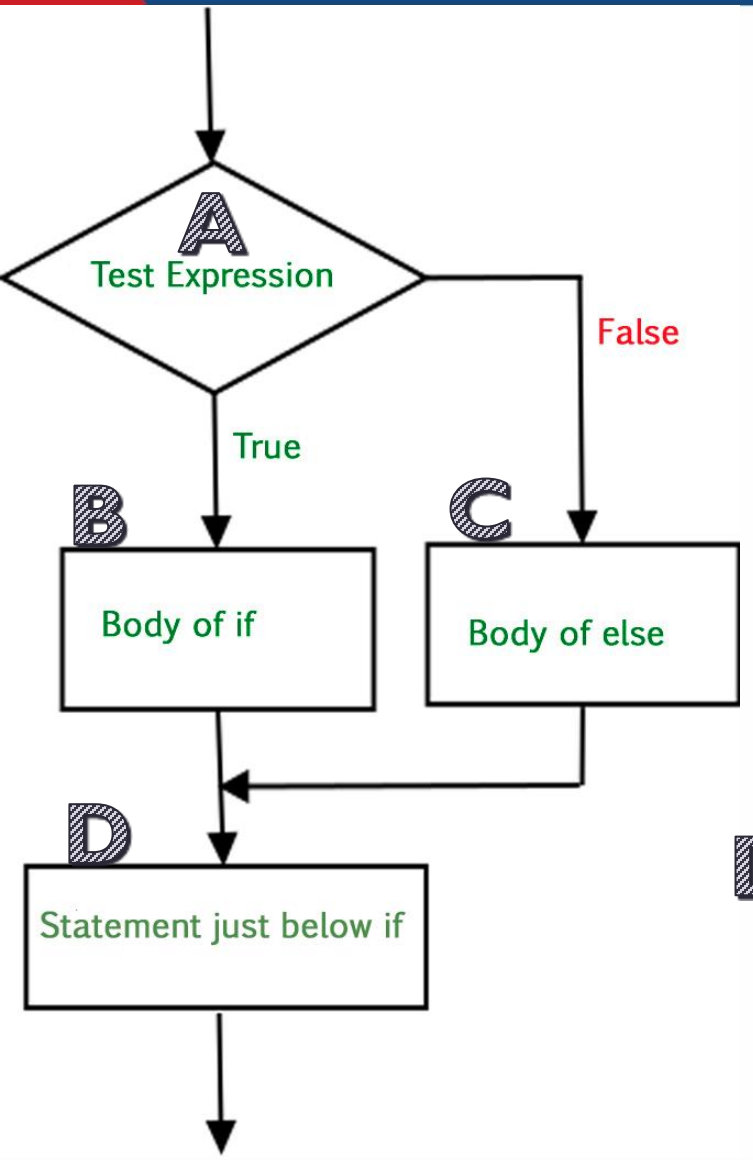
CONTROL FLOW: IF-ELSE

Flow: program only takes on path during an execution



[source]

IF-ELSE EXAMPLE




```
score = 35
if (score < 50):
    print ("Sorry, your score is less than 50")
    print ("[This is printed from the if Block]")
else:
    print ("Congratulations, your score is above 50")
    print ("[This is printed from the else Block]")
print ("[This is printed outside if-else Block]")
```

A B C D

What is the output?

WHAT IS THE OUTPUT?



```
number = input("Enter a number: ") 4

if number > 0:
    print("The number is positive.")
else
    print("The number is not positive.")
```

CLASS PRACTICE

A

```
a = 0
if a < 1:
    a += 1
    print(a)
else:
    a += 2
    print(a)
```

B

```
a = 0
if a >= 1:
    a += 1
    print(a)
else:
    a += 2
print(a)
```

C

```
a = 0
if a >= 1:
    a += 1
    print(a)
else:
    a += 1
a += 2
print(a)
```

Output:

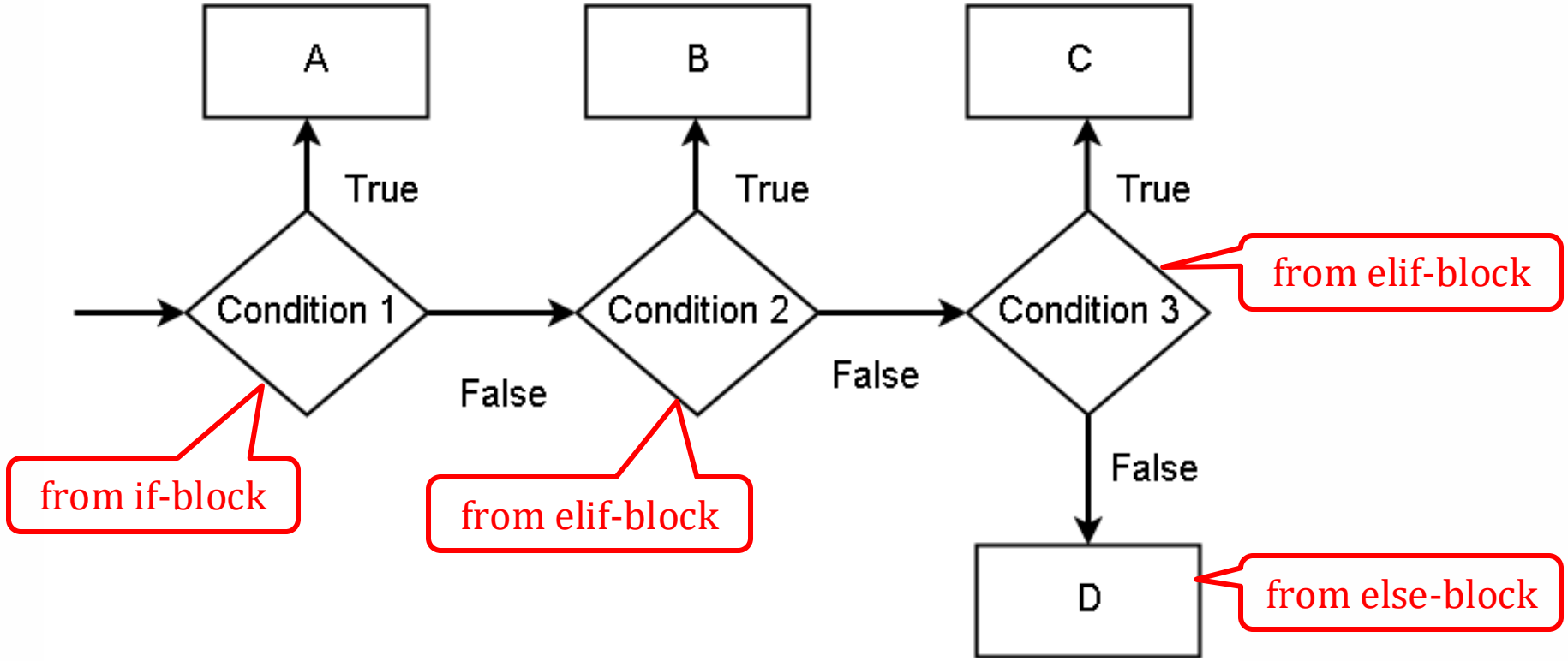
CONDITIONALS: IF-ELIF-ELSE-STATEMENTS

- Format:

```
if<boolean-expression>:
    <statement>
    …
elif<boolean-expression>:
    <statement>
    …
else:
    <statement>
    …
```

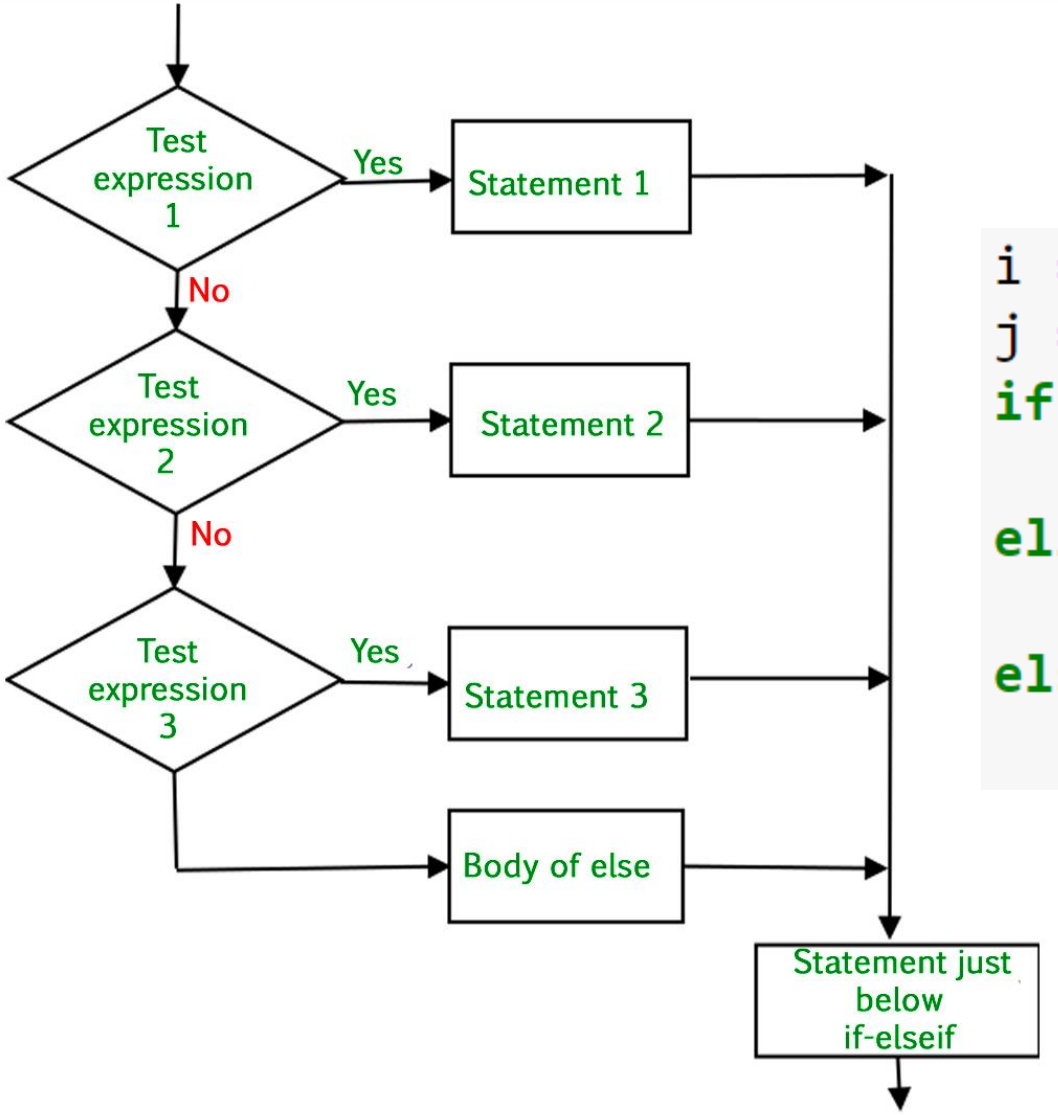
If the previous conditions were not true, then try this condition!

IF-ELIF-ELSE LOGIC



[source]

FLOW CHART: IF-ELIF-ELSE




```
i = 20
j = 30
if (i == j):
    print ("i is equal to j")
elif (i > j):
    print ("i is greater than j")
else:
    print ("i is less than j")
```

[source]

WHAT IS THE PURPOSE OF THIS PROGRAM?

```
number = int(input("Enter a number: "))
if number > 0:
    print("The number is positive!")
    if number % 2 == 0:
        print("The number is even!")
    else:
        print("The number is odd!")
elif number < 0:
    print("The number is negative!")
else:
    print("The number is zero!")
```

CLASS PRACTICE

Sample Code:

```
if a == 1:
    a += 2
    print(a)
elif a == 2:
    a += 3
    print(a)
elif a > 3:
    a += 4
    print(a)
else:
    a += 4
    print(a)
```

Output:

```
a = 0



a = 8



a = -5
```

COMMON MISTAKES (1/4)

```
ks_age = 20
if ks_age <30:
    print("You are young!")
elif ks_age <25:
    print("You are super young!")
else:
    print("You are still young!")
```

Output?

COMMON MISTAKES (2/4)

```
ks_age = 30
if ks_age <20:
    print("You are super young!")
elif ks_age < 30:
    print("You are young!")
else ks_age > 30:
    print("You are still young!")
```

Output?

COMMON MISTAKES (3/4)

```
ks_age = 30
if ks_age <20:
    print("You are super young!")
elif ks_age = 30:
    print("You are young!")
else:
    print("You are still young!")
```

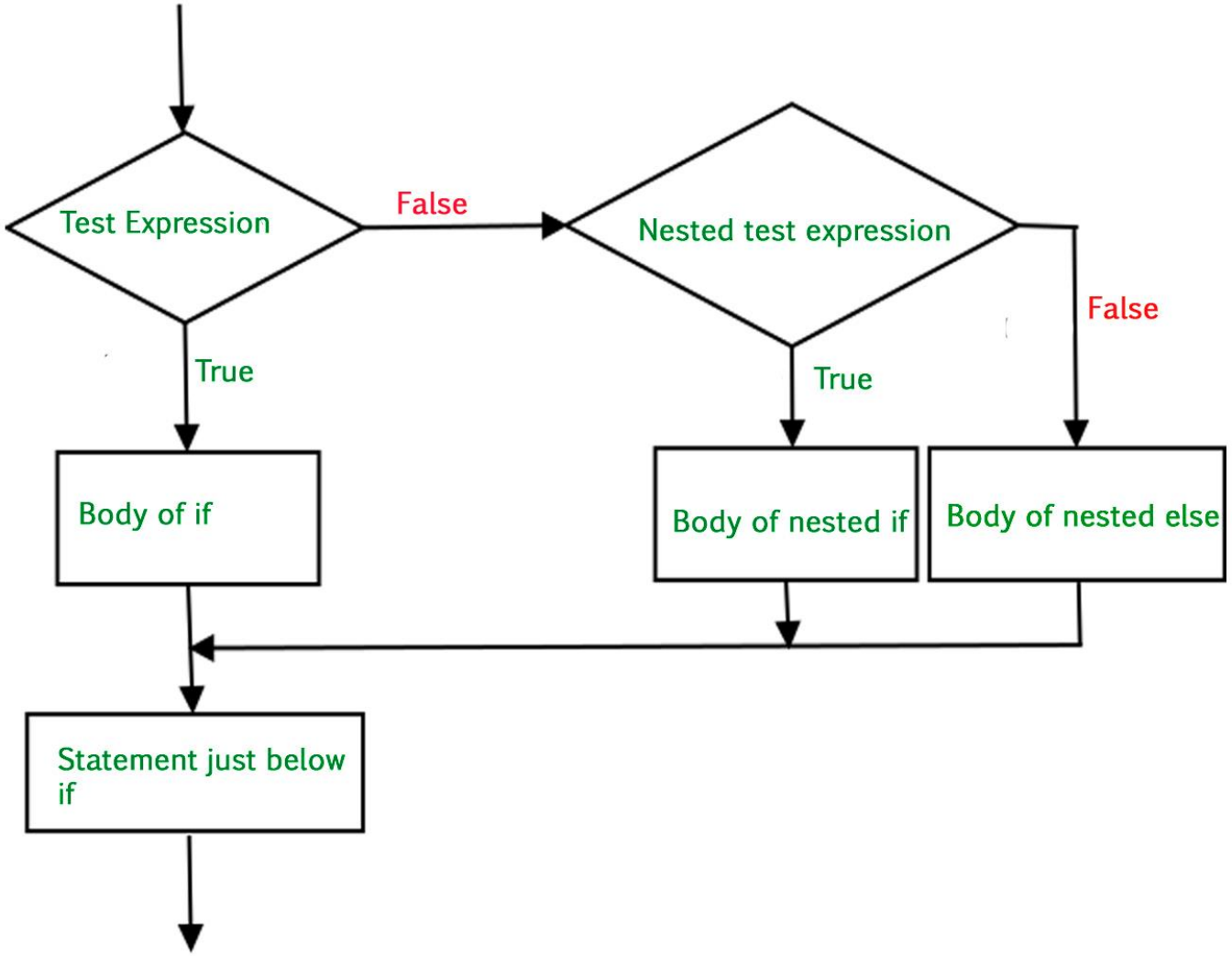
Output?

COMMON MISTAKES (4/4)

```
ks_age = 30
if ks_age <20:
    print("You are super young!")
    elif ks_age == 30:
        print("You are young!")
else:
    print("You are still young!")
```

Output?

CONTROL- FLOW: NESTED-IF-ELSE





[source]

CONTROL- FLOW: NESTED-IF-ELSE

```
if condition1:
    if condition2:
        # executed when both condition1 and condition2 are True
        statement1
    else:
        # executed when condition1 is True but condition2 is False
        statement2
else:
    # executed when condition1 is False
    statement3
```

NESTED IF-ELSE – EXAMPLE 1

```
temperature = 45
is_raining = True
is_hungry = False
if (temperature <= 25):
    if (is_raining and is_hungry):
        print ("Bring an umbrella to the restaurant.")
    else:
        print ("Play an online game!")
else:
    if (not is_raining or is_hungry):
        print("Go to the restaurant!")
    else:
        print ("Order foods delivery!")
```

NESTED IF-ELSE – EXAMPLE 2

speed = 100

```
if speed < 300:
    print ("The measured car speed is less than 300km/h.")
    if speed >= 250:
        print ("Dangerous!")
    elif speed >= 120 and speed <= 200:
        print ("Overspeed!")
    elif speed < 120:
        print ("Normal Speed!")
else:
    print ("Could not measure your car speed!")

print ("Good Luck!")
```

NESTED IF-ELSE – EXAMPLE 3

```
my_age = 27

if my_age <= 20:
    print("You are young")
    if my_age == 40 or my_age >17:
        print("You are not too young")
    else:
        print("Forever Young")
elif my_age > 10:
    if my_age <100 and my_age <= 16:
        print("You are too young?")
    else:
        print("Okay. You are young")
else:
    print("You are still young")
if my_age <100:
    print("Okay. I got it")
```

CLASS PRACTICE

Write a code (`nested if`) to check if an input is larger or less than 100. If the number is larger than 100, print out the result. Otherwise, check if it is an odd or even number. If it is an odd number, compute the value of log base 10 of it.

THANK YOU!